



Real-Time Surveillance System for Detecting Unauthorized Garbage Dumping Using Pose Estimation and LSTM

Guided by,

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Outline

- Introduction
- Objectives
- Workflow Diagram
- System Design
- Result Analysis (max 3 slides)
- Remaining work & GitHub link
- Reference

Introduction

Problem Statement:

- Urban areas face serious challenges due to unauthorized garbage dumping.
- Traditional CCTV systems only record footage and require manual monitoring.
- Existing automated systems focus on object detection rather than understanding human behavior.

This project proposes a **behavior-based intelligent surveillance system** that uses:

- **Pose Estimation (MediaPipe)**
- **Temporal Modeling (LSTM)**
- **Real-time Action Classification**

The system analyzes human motion patterns over time to detect illegal dumping activities accurately.

Objectives

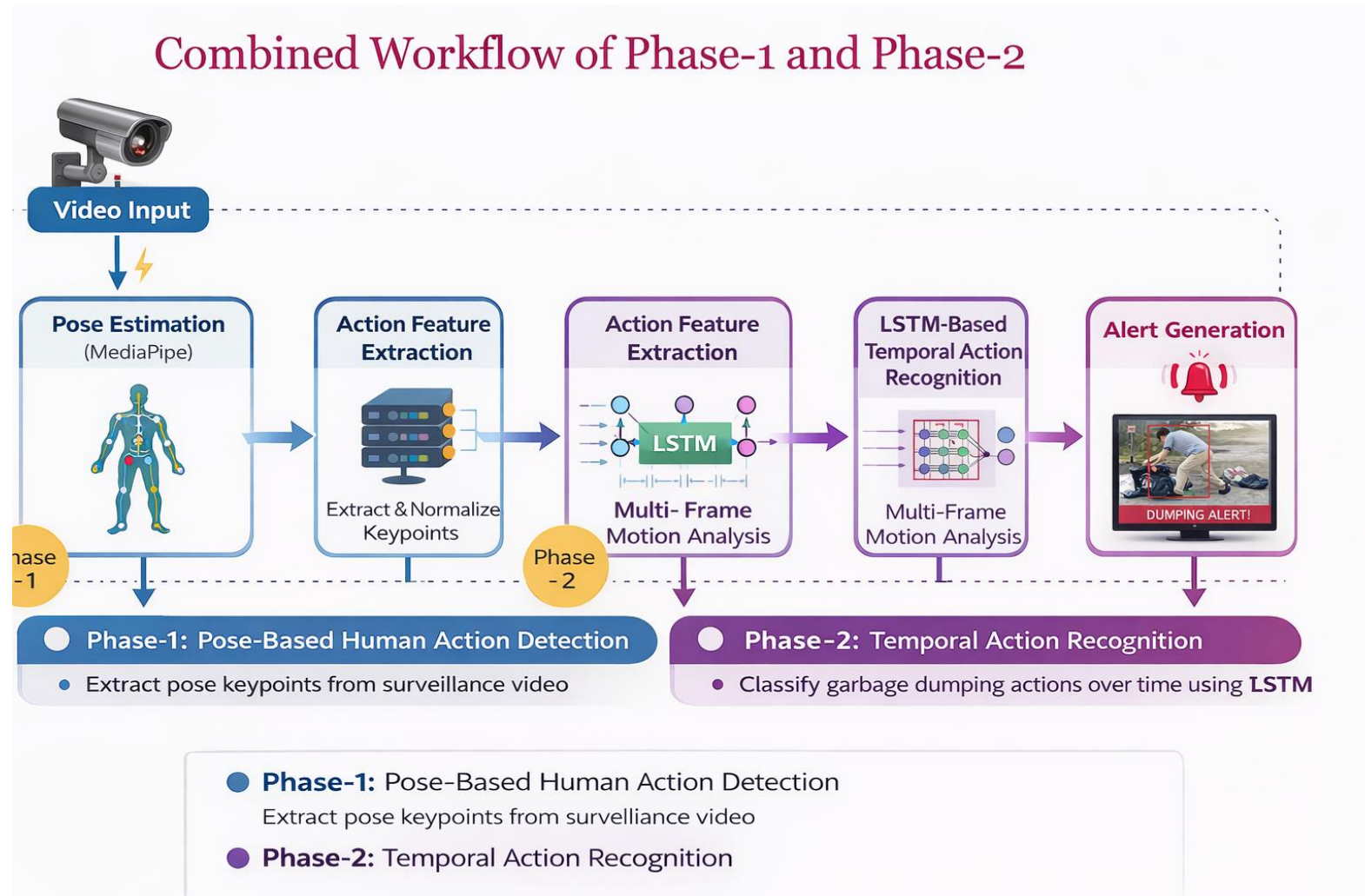
• S8 Objectives

- LSTM-Based Temporal Action Recognition for Garbage Dumping Detection
- Multi-Frame Motion Pattern Modeling for Improved Accuracy
- Robust Real-Time Surveillance Pipeline Integration
- Quantitative Performance Evaluation using Precision, Recall & F1-Score
- Research-Oriented Validation and Publication Preparation

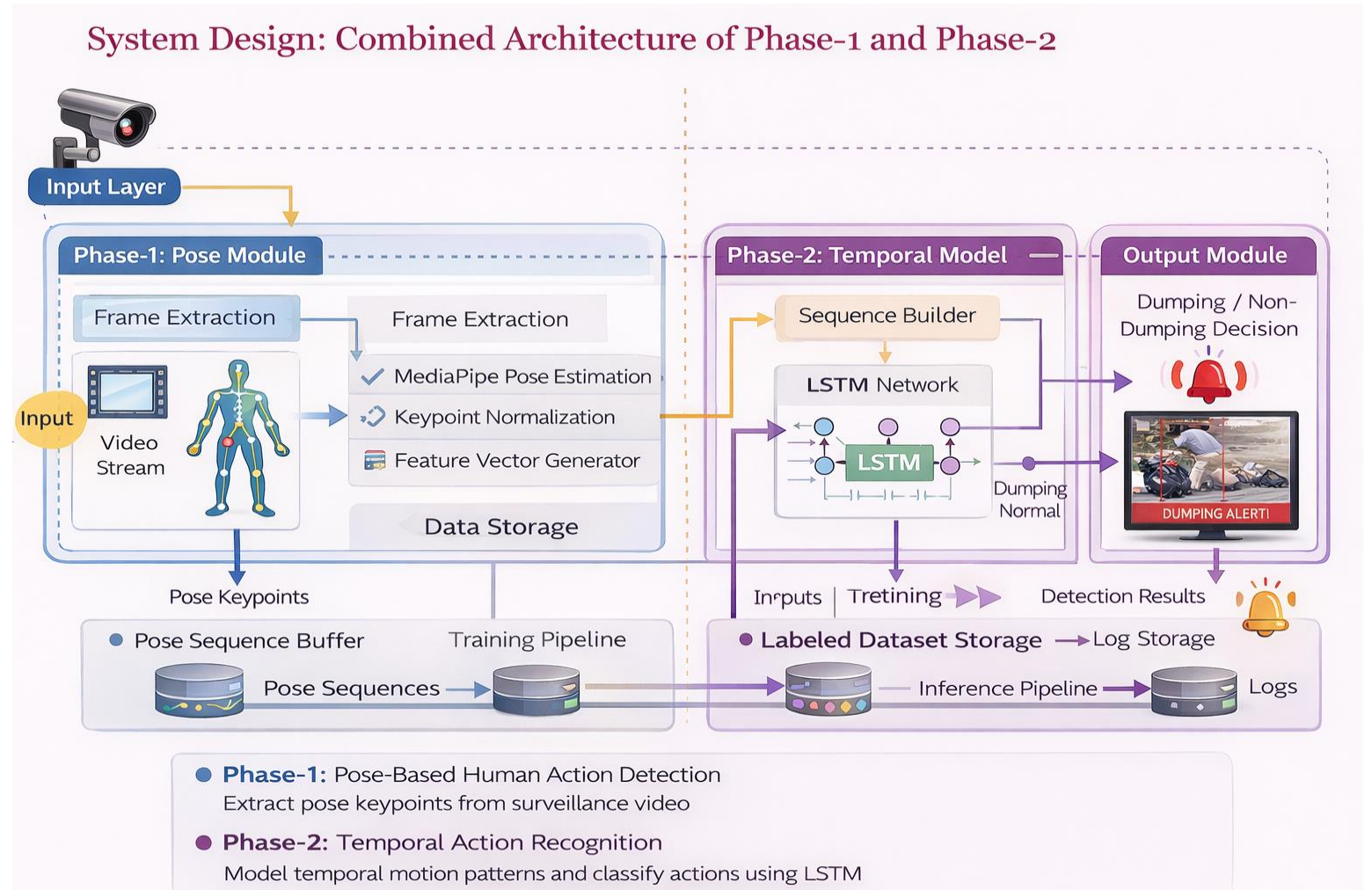
• S7 Objectives

- Real-Time Human Pose Estimation using MediaPipe
- Skeleton-Based Action Feature Extraction
- Detection of Bending, Carrying and Dropping Actions
- Intelligent Alert Generation for Unauthorized Dumping
- Prototype Development for Smart Surveillance

Workflow Diagram



Design Diagram



Result Analysis

Experimental Setup & Dataset

◆ Experimental Setup

- Language: Python
- Framework: TensorFlow / Keras
- Pose Extraction: MediaPipe
- Sequence Length: 34
- Classification Type: Binary
(Dumping vs Non-Dumping)

◆ Dataset Details

- Custom-collected surveillance videos
- Classes:
 - Class 0 → Normal Activity
 - Class 1 → Garbage Dumping
- Data Preprocessing:
 - Frame extraction
 - Pose keypoint normalization
 - Sequence generation
- Train-Test Split: (e.g., 80:20)

Why Custom Dataset?

- No public dataset specifically for illegal garbage dumping
- Real-world surveillance simulation
- Privacy preserved (only skeletal keypoints used)

Result Analysis

Performance Metrics

◆ Evaluation Metrics Used

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

◆ Current Results (Based on Given Metrics)

◆ Confusion Matrix Interpretation

- Model correctly identifies most normal actions.
- Limited detection of dumping class.
- Class imbalance affects performance.

Metric	Value
Accuracy	86%
Precision (Class 0)	0.86
Recall (Class 0)	1.00
F1-score (Class 0)	0.92
Class 1 Recall	0.00 (needs improvement)

Result Analysis

- **Performance Analysis & Observations**

- ◆ **Strengths**

- High accuracy for normal activity detection
 - Temporal modeling reduces random false alarms
 - Real-time inference achievable

- ◆ **Limitations**

- Small dataset size
 - Class imbalance problem
 - Occlusion and lighting affect pose extraction

- ◆ **Improvement Plan**

- Increase dumping samples
 - Apply data augmentation
 - Tune LSTM hyperparameters
 - Explore GRU / ConvLSTM comparison
 - Balance dataset

Remaining work

Item	Time required	Completion date(tentative)
Module name		
Module name		
• • •		
Paper writing		

GitHub link

Provide the github link to the project repository

Reference